

Why Indian Startups Fail: A Data-Driven Analysis

Startup Survival Outcomes — Python Analysis Report

Dataset: 62 Indian startups (post-deduplication) | **Analysis:** Python + SciPy

Status split: 38 Active (61.3%) · 24 Closed (38.7%)

Important: This is a manually curated sample. All findings represent patterns within this dataset and should not be generalised as population-level conclusions about the Indian startup ecosystem.

1. Data Cleaning & Preparation

Step	Action	Result
Duplicates removed	CoinSwitch, CoinDCX, BlackBuck, Livspace — kept first occurrence	66 → 62 rows
Column normalisation	Stripped whitespace, lowercased all string fields	Consistent values
Funding encoding	Ordinal mapping: Very Low=1, Low=2, Medium=3, High=4, Very High=5	New <code>funding_score</code> column
Null check	Zero nulls across all columns	Clean dataset

All percentages, statistics, and charts in this report are based on the 62-startup post-deduplication dataset.

2. Exploratory Data Analysis

2.1 Dataset Composition

- **62 startups** across 10 industries, founded between 2012–2023
- **Fintech** is the largest sector (n=15, 24.2%), followed by **E-commerce** (n=14, 22.6%)
- **High funding** is the most common funding level (n=23, 37.1%), while **Very Low** is the rarest (n=2)
- **20 unicorns** exist in the dataset — all concentrated in the High or Very High funding categories
- Of the 38 active startups, 20 are unicorns — a unicorn rate of **52.6% among active firms within this sample**

2.2 Years of Operation

Status	n	Mean	Median	Std	Min	Max
Active	38	7.61 yrs	8.0 yrs	1.99	4	10
Closed	24	5.29 yrs	4.5 yrs	3.21	2	13

Active startups had accumulated more years of operation at the time of analysis than closed startups. However, because active firms are still operating, these values should not be interpreted as final lifespan comparisons. The figures for active startups are right-censored — their eventual operating lifespans remain unknown.

The wider standard deviation for closed startups (3.21 vs. 1.99) indicates more variable timing of failure: some collapsed within 2 years, others after more than a decade.

3. Statistical Analysis

3.1 Point-Biserial Correlation — Funding Score vs. Active Status

$$r = 0.49, p < 0.001$$

A moderate-to-strong positive correlation: higher funding is associated with a greater likelihood of being active. This is statistically significant. In practical terms, each step up the funding ladder is associated with improved survival probability within this sample. However, $r = 0.49$ also indicates that roughly half of the variance in survival outcomes is explained by factors beyond funding level alone — including industry, execution quality, and market timing.

3.2 Chi-Square Test — Funding Range vs. Status

$$\chi^2 = 24.16, p = 0.0001, df = 4 \mid \text{Cramér's } V = 0.62$$

The association between funding range and startup status is highly statistically significant, with a large effect size (Cramér's $V > 0.50$ is generally considered strong). This is the most robust statistical finding in the analysis.

Funding Range	Active	Closed	Failure Rate
Very Low (n=2)	0	2	100.0%
Low (n=9)	0	9	100.0%
Medium (n=15)	9	6	40.0%
High (n=23)	20	3	13.0%
Very High (n=13)	9	4	30.8%

The Very Low and Low findings (100% failure) are striking but based on small cell sizes (n=2 and n=9 respectively). The directional pattern is consistent with the hypothesis, but these rates should be

treated with caution rather than as definitive estimates.

The Very High funding failure rate of 30.8% — higher than the High category's 13.0% — is discussed further in the outlier section.

3.3 Mann-Whitney U Test — Survival Years by Status

$U = 674.5, p = 0.0015$

A statistically significant difference in years of operation was observed between active and closed startups. This non-parametric test is appropriate given the sample size ($n=62$) and the non-normal distribution of survival years. As noted above, the result reflects accumulated operating years at the time of analysis, not final lifespan — active startups are still running and subject to right-censoring.

3.4 Kruskal-Wallis Test — Survival Years across Industries

$H = 8.95, p = 0.44$

No statistically significant differences in survival years were observed across industries. However, this result should not be interpreted as evidence that industry has no effect on startup success or failure. The test was applied to survival years — a continuous measure of operating duration — rather than to the binary outcome of success versus failure. Given the small and uneven sample sizes across industry groups, the test also has limited power to detect differences even if they exist.

3.5 Funding Stage vs. Failure Rate

Stage	n	Failure Rate
Seed	9	100.0%
Series A	8	87.5%
Series B	10	30.0%
Series C+	35	14.3%

Failure rates decline substantially as startups progress through funding stages. The drop from Series A (87.5%) to Series B (30.0%) is particularly pronounced, suggesting that within this sample, reaching Series B may represent a meaningful survival threshold — likely reflecting demonstrated product-market fit or operational viability. This pattern is consistent across the dataset, though the small sample sizes at early stages (Seed $n=9$, Series A $n=8$) warrant caution in generalisation.

4. Industry-Level Analysis

Industry	n	Closed	Failure Rate	Unicorns
Cleantech	1	1	— excluded	0
Healthtech	4	3	75.0% $\triangle$	0
Edtech	5	3	60.0%	2
Mobility	2	1	50.0%	1
Logistics	4	2	50.0%	2
Agritech	5	2	40.0%	0
SaaS	8	3	37.5%	1
Fintech	15	5	33.3%	5
Spacetech	4	1	25.0%	0
E-commerce	14	3	21.4%	9

Cleantech was excluded from interpretation due to insufficient sample size (n=1). A single observation cannot support any meaningful conclusion about the sector.

E-commerce exhibited the lowest failure rate (21.4%) and the highest unicorn concentration within the sample. While this suggests relatively strong survival outcomes, the finding should be interpreted as a pattern within the dataset rather than evidence of sector-wide superiority.

Healthtech displayed the highest failure rate within the sample (75%), although the sample size was limited (n=4). This finding should therefore be interpreted cautiously. The result may reflect genuine structural challenges in the sector — such as longer product development cycles and regulatory complexity — but it cannot be treated as a definitive characterisation of Healthtech outcomes.

Fintech remains the largest sector in the sample (n=15) and generated both the highest number of failures (5) and the most unicorns (5 excluding E-commerce). This makes it the most competitive and internally diverse sector observed.

Spacetech had a lower-than-expected failure rate (25%), but again the sample is too small (n=4) to draw reliable conclusions.

5. Outlier Analysis

5.1 High-Funded Startups That Failed

Startup	Funding	Stage	Yrs of Operation	Observation
Builder.ai	Very High	Series C+	13	Longest-operating startup before closure. Raised substantially but faced reported

Startup	Funding	Stage	Yrs of Operation	Observation
				accounting irregularities.
Dunzo	Very High	Series C+	10	Extensively funded but unit economics did not scale.
The Good Glamm Group	Very High	Series C+	10	Aggressive acquisition strategy without sufficient profitability.
BlueSmart	Very High	Series B	6	EV mobility sector; operational and regulatory challenges despite high capital.
Log9	High	Series B	10	Cleantech; extended runway but did not achieve commercial scale.
Otipy	High	Series B	5	E-commerce; failed despite reaching Series B.
Astra	High	Seed	2	Spacetech; shortest-operating high-funded startup. Deep-tech burn rates can be severe even at seed stage.

These seven cases demonstrate that high capital access does not guarantee survival. All reached advanced or well-resourced stages before closing. Their failures point to execution challenges, governance issues, or business model weaknesses rather than capital scarcity. This nuance is one of the most analytically meaningful findings in the dataset.

5.2 Notable Survivor

Physics Wallah achieved unicorn status with only High funding at Series B — one of very few unicorns in the dataset that did not require Series C+ funding. This suggests that in certain B2C models with strong organic demand, capital efficiency can substitute for large consecutive funding rounds. It is an outlier in the positive direction.

5.3 Right-Censoring Note on "Active Survivors"

All 38 active startups are still operating at the time of analysis. Their current years of operation do

not represent final survival durations. Any comparison of operating years between active and closed groups must account for this asymmetry.

6. Statistical & Analytical Insights

These insights emerge specifically from statistical analysis and cannot be easily extracted from static pivot tables:

- The Series B threshold effect.** Failure rates decline from 87.5% (Series A) to 30.0% (Series B). The non-linear drop at this stage suggests a discrete inflection point in startup viability. A pivot table can show counts; only quantified rates reveal the magnitude of the cliff.
 - Very High funding exhibits a higher failure rate than High funding (30.8% vs 13.0%).** This counterintuitive finding suggests that very large funding rounds do not uniformly increase survival probability. Some Very High funded startups in this sample appear to have raised capital at scale without resolving underlying business model challenges.
 - Active startups show a compressed operating year distribution (std=1.99)** while closed startups show wide variance (std=3.21). Startups that remained active tended to cluster in a narrow band of 6–10 years of operation. Closures were unpredictable in timing — from 2 years to 13 years. This asymmetry has implications for risk assessment.
 - No unicorns in this dataset originated from the Low or Very Low funding categories.** However, this observation is limited to the sample and should not be generalised to the broader startup ecosystem. It may reflect selection effects in how this dataset was curated.
 - Founding year is not a significant differentiator.** Active startups had a median founding year of 2017; closed startups had a median of 2019. The distributions are close, indicating that recency bias (newer startups having less time to succeed) is not a major confound in this dataset.
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7. Hypothesis Evaluation

Hypothesis: *Funding level and industry characteristics are associated with startup survival outcomes in India, with lower-funded startups exhibiting higher failure rates than heavily funded startups.*

Verdict: PARTIALLY SUPPORTED

Component	Evidence	Verdict
Lower funding → higher failure	Very Low/Low = 100% failure rate in sample; $\chi^2=24.16$, $p=0.0001$; $r=0.49$	Supported (with caveat on small cell sizes)
Heavily funded = better survival	High funding: 13.0% failure rate	Supported
Heavily funded = guaranteed survival	Very High: 30.8% failure; 5 high-funded closures observed	Not supported

Component	Evidence	Verdict
Industry characteristics matter	Failure rates vary 21–75% across sectors (descriptive)	Directionally supported
Industry effect is statistically significant	Kruskal-Wallis $p=0.44$ on survival years; small sector sample sizes	Not statistically confirmed

The hypothesis captures the directional pattern accurately. However, it overstates the protective power of high funding and does not account for the complexity of how industry and funding interact. The more precise finding is that funding level is a necessary but not sufficient condition for survival, and that industry-level patterns, while descriptively visible, are not statistically confirmed by formal testing in this sample.

8. Key Findings Summary

- Funding level is the strongest measurable predictor of survival in this sample** — confirmed by point-biserial correlation ($r=0.49$, $p<0.001$), chi-square ($\chi^2=24.16$, $p=0.0001$), and Cramér's V (0.62 — large effect size).
- Very Low and Low funded startups had a 100% failure rate in this dataset.** These findings are directionally consistent with the hypothesis but based on small cell sizes ($n=2$ and $n=9$) and should not be extrapolated.
- Reaching Series B is the most pronounced survival milestone in this sample** — failure rate falls from 87.5% (Series A) to 30.0% (Series B) to 14.3% (Series C+).
- Very High funding does not guarantee survival** — 30.8% of Very High funded startups closed, including three that operated for 10 years or more. While High-funded startups exhibited lower failure rates than Very High-funded startups in this sample, several heavily funded failures indicate that capital alone does not guarantee startup survival.
- E-commerce exhibited the strongest survival outcomes within the sample**, while Healthtech recorded the highest observed failure rate (75%), although Healthtech's sample size was limited ($n=4$).
- Active startups had accumulated significantly more operating years than closed startups** at the time of analysis (medians: 8 vs. 4.5 years; Mann-Whitney $p=0.0015$). Because active firms are still operating, these figures represent right-censored values and should not be treated as final lifespan estimates.
- All 20 unicorns in this dataset originated from the High or Very High funding categories.** This pattern holds consistently across the sample, though it should not be generalised beyond it.
- Industry-level failure rate differences are descriptively visible but not statistically confirmed** by formal testing. Sector-level findings should be treated as directional observations.

9. Business Implications & Recommendations

For Investors

- **Stage progression is a stronger filter than funding amount alone.** A startup reaching Series B has demonstrated something that capital alone cannot buy — evidence of product-market fit or operational resilience.
- **High funding range outperformed Very High in survival rate (13.0% vs 30.8% failure).** Disciplined capital deployment appears to correlate with better outcomes than maximum fundraising in this sample.
- **E-commerce and Fintech accounted for the majority of unicorns** in this dataset. Portfolio construction aimed at large-scale outcomes may benefit from weighting these sectors, while acknowledging that this reflects sample composition rather than a definitive market signal.
- **Exercise caution when evaluating Healthtech startups in this sample**, as the sector displayed a high observed failure rate. However, conclusions are constrained by the limited number of observations (n=4) and should not be treated as a blanket sector verdict.

For Founders

- **Reaching Series B should be treated as the first genuine survival milestone.** The data suggests this stage reflects real external validation — investors willing to commit at Series B have assessed product and market fit, not just the founding team.
- **Sustained high capital is not a substitute for unit economics.** Builder.ai, Dunzo, and The Good Glamm Group each raised at scale and still closed. The common thread in these failures appears to be business model or governance weaknesses that capital could not resolve.
- **Industry-level patterns suggest differing survival outcomes across sectors.** However, due to limited sample sizes in some industries, these results should be treated as directional rather than definitive guidance for sector selection.

For Business Analysts and Stakeholders

- **Funding stage captures more signal than funding range alone.** A Series C+ startup in the Medium funding range has substantially better survival odds than a Series A startup in the High range. Stage reflects milestone achievement; range reflects capital volume.
- **The 4–5 year mark represents an observed risk window** — the median closure occurred at 4.5 years of operation. Startups that cannot demonstrate operational viability within this window appear to face elevated risk, based on patterns in this dataset.

10. Limitations

- The dataset represents a curated sample of 62 Indian startups and is not intended to represent the entire Indian startup ecosystem.

- Certain industries contain relatively small sample sizes, which may affect the reliability of industry-level comparisons.
- Active startups are still operating at the time of analysis. Therefore, years of operation for active startups represent accumulated years to date rather than final lifespan.
- Findings should be interpreted as observed patterns within the sample and not as causal relationships.

Report version 3.0 — revised per analytical review. All findings based on 62-startup post-deduplication dataset. Cleantech excluded from industry interpretations (n=1). Active startup operating years treated as right-censored values throughout. Industry-level findings are descriptive patterns within this sample only.